TMDB Movie Data Analysis

Key findings from 18 blockbuster films, using data pulled from the TMDB API

> **The short version.** Every movie in this set made several times its budget back, so this is really a study of how big hits compare to *other* big hits. Within that group, a bigger budget didn't reliably mean more money, the most common genres (Action and Sci-Fi) had the lowest typical returns, and popularity and rating moved together fairly strongly. Franchise and standalone movies looked surprisingly similar, but with only two standalone films, that comparison can't carry much weight.

How I got here

I fetched 19 movie IDs from the TMDB API (one, ID 0, doesn't exist and was skipped), including cast and crew. Then I cleaned the data: flattened the nested JSON fields, converted budgets and revenues to millions of US dollars, treated zeros and placeholder text as missing, and kept only released movies. From there I calculated profit (revenue minus budget) and ROI (revenue divided by budget), ranked the movies, compared franchises with standalone films, and looked at directors. All the code is in the accompanying Jupyter notebook.

Major findings

1. Every movie was a big earner, and budget didn't decide by how much

No movie came anywhere near the break-even line. The weakest return was *Star Wars: The Last Jedi* at roughly 4.4 times its budget, and the strongest was *Avatar* at over 12 times. Spending more didn't guarantee earning more: *The Last Jedi* cost about $300M and made around $1.33B, while *Harry Potter and the Deathly Hallows: Part 2* cost about $125M and made roughly the same. At the very top, *Avatar* and *Avengers: Endgame* both earned close to $2.9B, but *Endgame* needed about $120M more to get there.

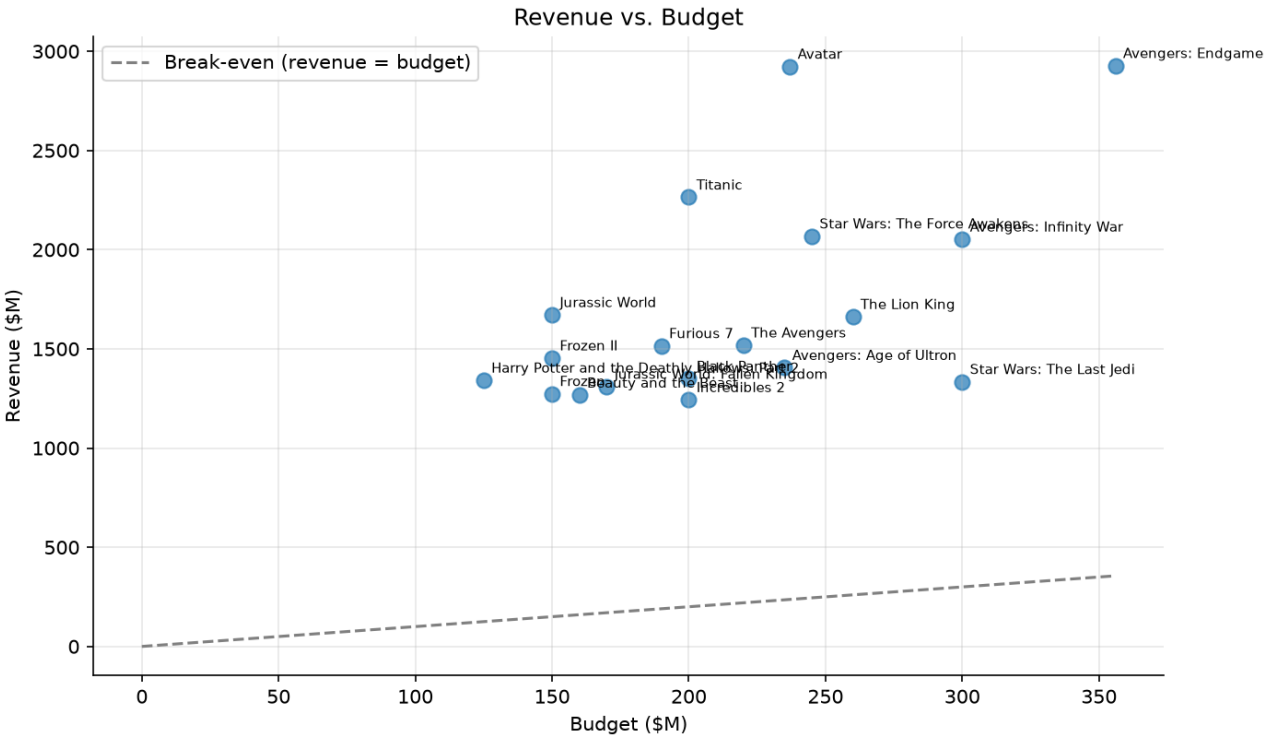


Figure 1. Revenue vs. budget. The dashed line is break-even; every movie sits far above it.

2. The most common genres had the lowest typical returns

Action and Science Fiction appear in 10 movies each, yet they have the lowest median ROI, around 7x. Their boxes are also pulled out by one huge outlier on the right, which is *Avatar*. Romance, Fantasy and Drama sit higher, but each covers only two to four movies, so I wouldn't read too much into their position. A fair way to put it: in this group, action-heavy spectacle tends to cost so much that its returns end up ordinary compared with other hits.

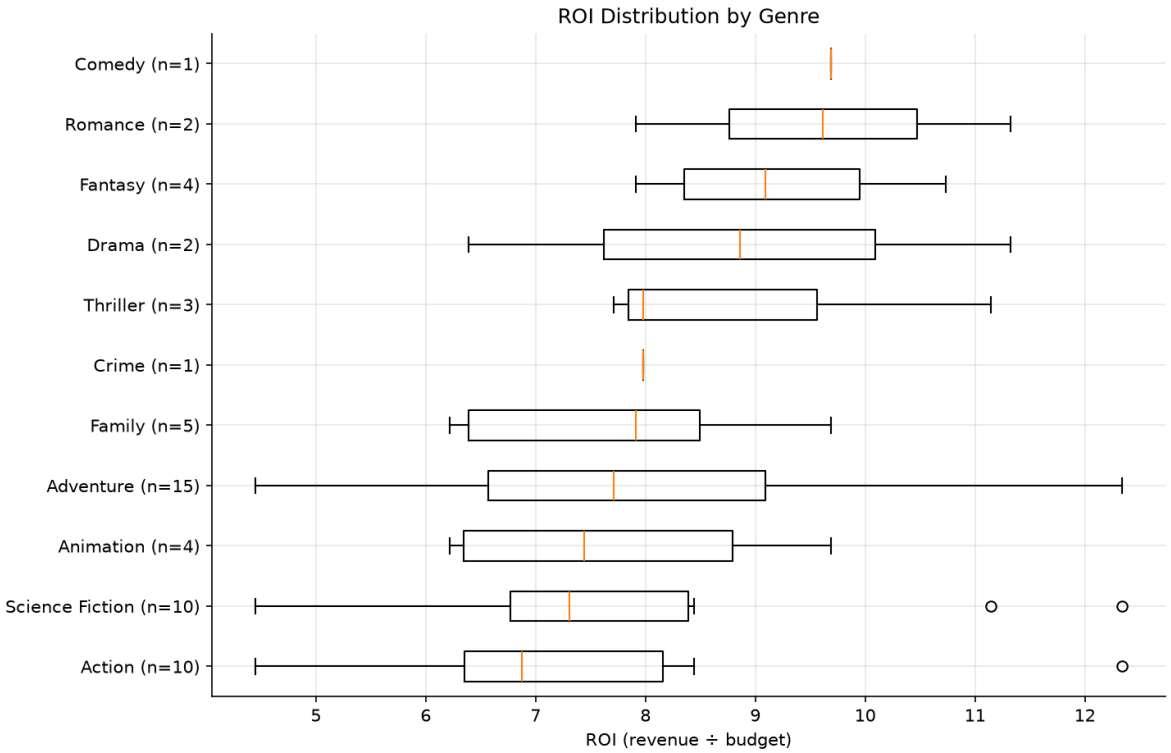


Figure 2. ROI by genre, sorted by median. n is the number of movies in each genre; a movie with several genres counts in each.

3. Popular movies also tended to be the best rated

Popularity and average rating have a correlation of 0.74, which is a fairly strong positive link. The top-right corner belongs to the Avengers: *Infinity War* and *Endgame* are both the most popular and the highest rated. Ratings overall sit in a narrow band, from about 6.5 to 8.25, so the differences in quality are smaller than the differences in popularity.

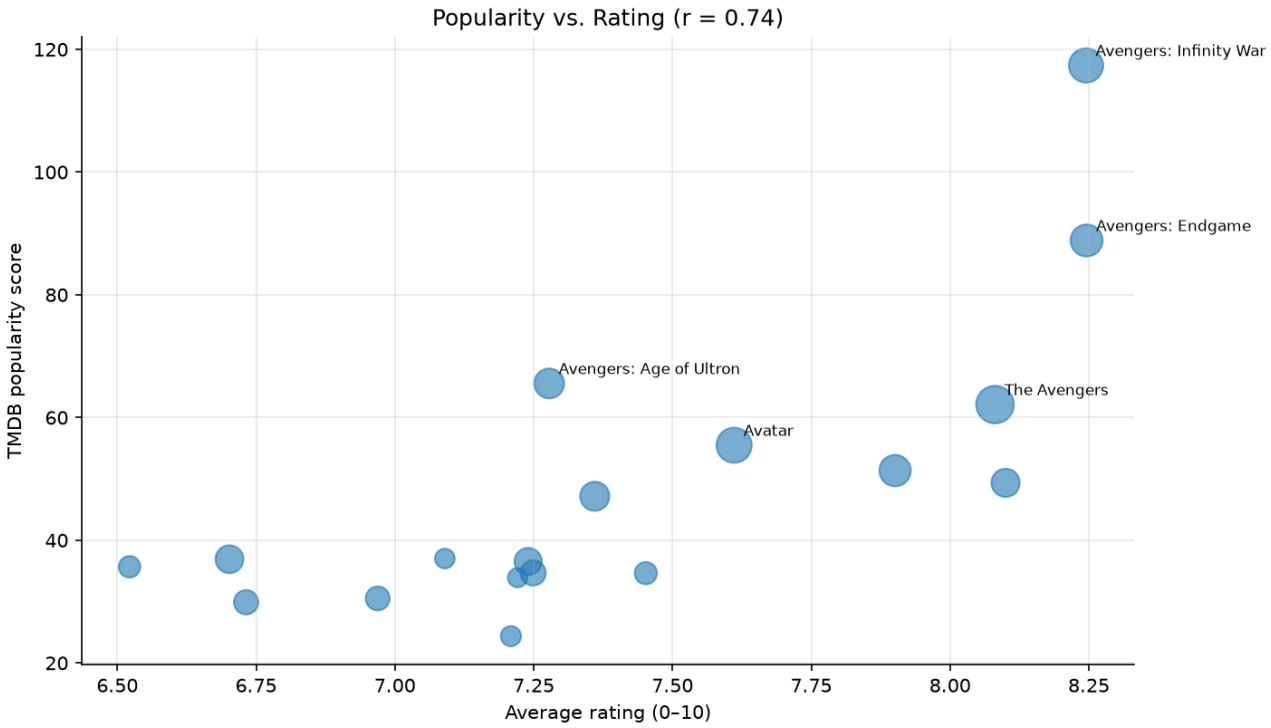


Figure 3. Popularity vs. average rating. Bigger dots mean more votes; the five most popular movies are labelled.

4. Strong years were mostly years with more of these movies

2015 had the highest total revenue, about $6.7B, but it also had four movies in the list. 2018 and 2019 come close for the same reason. Among single-movie years, 2009 (*Avatar*) stands out at about $2.9B. So this chart says more about which movies were picked for the project than about the film industry's yearly trend.

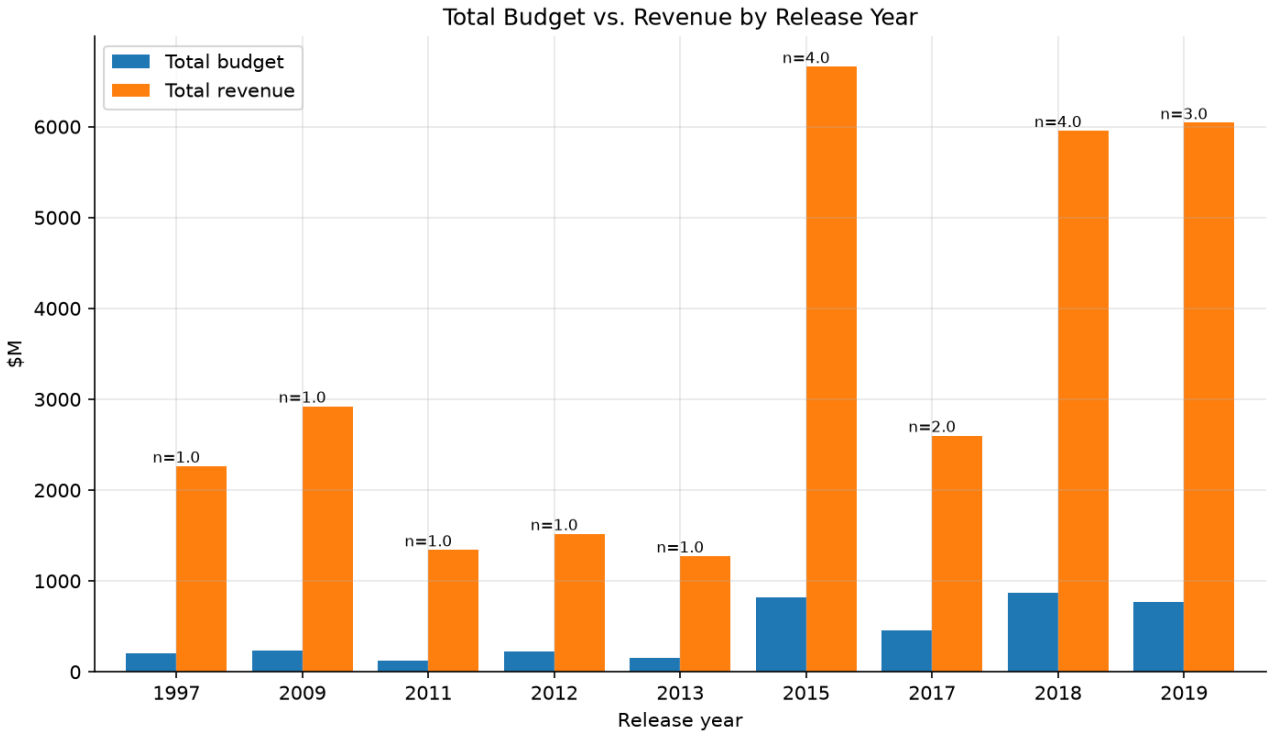


Figure 4. Total budget and revenue by release year. n is the number of movies released that year.

5. Franchise and standalone movies looked more alike than expected

16 of the 18 movies belong to a franchise. Franchise films had bigger budgets (about $218M vs. $180M on average) and higher popularity. The two standalone films earned slightly more on average, got a better median ROI (about 9.6x vs. 7.8x), and were rated almost exactly the same. Two movies is far too few to say standalone films do better. What I can say is that, in this dataset, being part of a franchise didn't come with an obvious payoff.

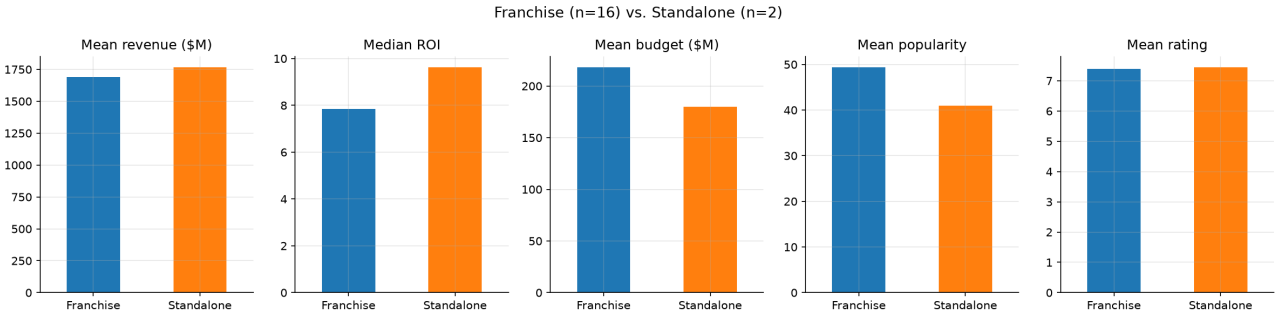


Figure 5. Franchise vs. standalone movies across five metrics.

Side findings

These weren't part of the brief, but they came up while I was working through the data.

- **Sequels don't automatically beat the original.** *The Last Jedi* had a bigger budget than *The Force Awakens* ($300M vs. about $245M) but earned roughly $700M less. *Jurassic World: Fallen Kingdom* also earned less than *Jurassic World*. *Frozen II* went the other way, beating *Frozen* on the same $150M budget, and the Avengers series dipped with *Age of Ultron* before climbing to *Endgame*.
- **There are no flops in this dataset.** The list is essentially a who's-who of the biggest box office hits, so "lowest profit" and "lowest ROI" really mean "least spectacular success." Any conclusion here is about winners only.
- **The Avengers carry the popularity numbers.** Four of the five most popular movies are Avengers films. Part of the 0.74 correlation is likely an Avengers effect, and it would be worth recalculating it without them.
- **The data leans heavily on the 2010s.** 16 of the 18 movies came out between 2011 and 2019. The only older films are *Titanic* (1997) and *Avatar* (2009).
- **Titanic's numbers are understated.** Revenues aren't adjusted for inflation, so *Titanic*'s 1997 earnings would look considerably bigger in today's money, and its position in the rankings would likely improve.

Limitations and next steps

The main limitation is the sample: 18 hand-picked blockbusters can show patterns among hits, but not what makes a movie a hit. TMDB's popularity score also reflects recent activity on the site, so it's a snapshot from when I fetched the data rather than a measure of all-time success. If I extended this project, I would fetch a much larger and more varied set of movies (including ones that lost money), adjust revenues for inflation, and test whether the genre and franchise patterns still hold.